

ADDITIONAL PROBLEM SHEET 2

Topics covered: Chapters 3–5.

1. Modified Lotka–Volterra (Exam section B type, but a little short)

Consider the following system:

$$\frac{dx}{dt} = r_1 x \left[1 - \frac{x}{k_1 + a_1 y} \right], \quad (1a)$$

$$\frac{dy}{dt} = r_2 y \left[1 - \frac{y}{k_2 + a_2 x} \right], \quad (1b)$$

where the constants $r_1, r_2, k_1, k_2, a_1, a_2$ are all assumed positive.

- (a) By setting $x = \hat{x}X$, $y = \hat{y}Y$ and $t = \hat{t}T$, show that the system can be written in the form

$$\frac{d\hat{x}}{d\hat{t}} = \hat{x} \left[1 - \frac{\hat{x}}{1 + \hat{y}} \right], \quad (2a)$$

$$\frac{d\hat{y}}{d\hat{t}} = \gamma \hat{y} \left[1 - \frac{\hat{y}}{\alpha + \beta \hat{x}} \right], \quad (2b)$$

and give the values of X, Y, T, α, β and γ in terms of the original parameters. This is one technique for simplifying such systems. We have reduced the equations from dependence on six parameters to just three.

- (b) Determine all the equilibria of this system and analyse their stability.
- (c) Part (b) should have told you that there are only stable equilibria for a subset of all the admissible parameters $r_1, r_2, k_1, k_2, a_1, a_2 > 0$. What choice of parameters can we make such that there is always a stable equilibrium? Does this change lead to a physically reasonable model? (Consider the permissible equilibria and what they mean in this analysis).

2. More population dynamics stability analysis (Exam between parts A and B)

Consider the following system of ordinary differential equations for the time-dependent interaction of two populations u and v :

$$\frac{du}{dt} = -(u - 1)(u - a) + \gamma uv, \quad (3a)$$

$$\frac{dv}{dt} = v(1 - v) - \gamma uv. \quad (3b)$$

We assume the interaction term γ is positive so that u is the predator, and that a is a real constant.

- (a) State a condition on u and v for the system to be considered physically reasonable.
- (b) Find the four possible equilibria of the system.
- (c) Show that the Jacobian $\mathbf{J}$ for eq. (3) takes the form

$$\mathbf{J} = \begin{pmatrix} -2u_0 + (1+a) + \gamma v_0 & \gamma u_0 \\ -\gamma v_0 & 1 - 2v_0 - \gamma u_0 \end{pmatrix}, \quad (4)$$

where u_0 and v_0 are equilibrium solutions to (3).

- (d) State the conditions on the parameter γ for the stability of the equilibria:
 - (i) $(u_0, v_0) = (1, 0)$,
 - (ii) $(u_0, v_0) = (a, 0)$.

3. ODE species interaction (Exam section A, 2017)

Consider the following system used to model herbivore–plankton interaction,

$$\frac{du}{dt} = r_1 u \left[(K - u) - \frac{Bv}{C + u} \right], \quad (5a)$$

$$\frac{dv}{dt} = r_2 v \left[\frac{u}{C + u} - Av \right], \quad (5b)$$

where r_1, r_2, A, B, C, K are positive constants.

- (a) Show that we can nondimensionalise this model so that it takes the form

$$\frac{d\hat{u}}{d\hat{t}} = \hat{u} \left[(k - \hat{u}) - \frac{\hat{v}}{1 + \hat{u}} \right], \quad (6a)$$

$$\frac{d\hat{v}}{d\hat{t}} = r\hat{v} \left[\frac{\hat{u}}{1 + \hat{u}} - a\hat{v} \right]. \quad (6b)$$

- (b) Describe the various inter-species interactions of the system and the intra-species interactions.
- (c) Assume that $r = 0$ and that $\hat{v}$ takes a constant value, $\hat{v}_0$. If $\hat{v}_0 \gg \hat{u}$ then to a reasonable approximation,

$$\frac{d\hat{u}}{d\hat{t}} = -\frac{\hat{u}\hat{v}_0}{1 + \hat{u}}. \quad (7)$$

Using this approximation, show that $\hat{u}$ satisfies

$$\log(\hat{u}) + \hat{u} = -\hat{v}_0 \hat{t} + D, \quad (8)$$

with D a constant of integration. Further, show that if $\hat{u} \ll 1$ when $\hat{t}$ gets large ($\hat{t} \gg 1$), then $\hat{u}$ decays exponentially.

4. Stability analysis of population interaction (Exam section B, 2017 resit)

Consider the following system of ODEs used to model herbivore–plankton interaction,

$$\frac{du}{dt} = u \left[(k - u) - \frac{v}{1 + u} \right], \quad (9a)$$

$$\frac{dv}{dt} = rv \left[\frac{u}{1 + u} - a \right], \quad (9b)$$

where $r, a \geq 0$ and $k \in \mathbb{R}$ are constants.

- (a) Find the three permissible equilibria of the system.
- (b) For what parameter ranges can we have physically valid equilibria for which both u_0 and v_0 are nonzero?
- (c) Assume $v(0) = 0$. Solve for $u(t)$ with an initial condition $u(0) = u_0$.
- (d) No longer assume $v(0) = 0$. Instead, assume $k < 0$. Perform a linear stability analysis for all physically permissible equilibria.

5. More population interaction (Exam 2021 resit)

Consider the following generalised predator–prey system,

$$\begin{aligned}\frac{dx}{dt} &= ax \left(1 - \frac{x}{K}\right) - byf(x), \\ \frac{dy}{dt} &= cyf(x) - dy,\end{aligned}$$

where $x(t)$ is the number of prey and $y(t)$ is the number of predators, both of which vary in time. The parameters a, b, c, d and K are positive constants. The function $f(x)$ is given by

$$f(x) = \frac{x}{x + E}$$

for some positive constant E .

- (a) Give a biological interpretation of the predator–prey interaction arising from this form of $f(x)$.
- (b) Choose an appropriate scaling for t, x and y , to show that the nondimensionalised system in $\hat{t}, \hat{x}$ and $\hat{y}$, once hats have been removed, is

$$\begin{aligned}\frac{dx}{dt} &= x(1 - x) - \frac{xy}{x + \delta}, \\ \frac{dy}{dt} &= y \left(\frac{\alpha x}{x + \delta} - \mu \right),\end{aligned}$$

stating the values of α, δ and μ .

- (c) Show that there are two equilibria of the system which always exist, and a further, (x^*, y^*) , which exists under conditions you should state.
- (d) Show that if (x^*, y^*) is an equilibrium, then the other two equilibria are unstable.
- (e) Show that (x^*, y^*) is stable if $(1 + \delta)\mu > \alpha(1 - \delta)$.

6. Nondimensionalisation and model simplifications (Exam 2022)

Consider the following model for a species $u(x, t)$, which we will see in chapter 6:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + ru \left(1 - \frac{u}{K}\right), \quad x \in [0, L], \quad (10)$$

where $D, r, K > 0$.

This is the Fisher–Kolmogorov equation modelling an invasive species, and you’ll notice it’s a partial differential equation: the population, u , is a function of space, x , and time, t . We’ll derive it chapter 6 and learn all about it then, so don’t panic. We’re going to use it for nondimensionalisation practice.

- (a) Just before we begin, if $r = 0$, what does this equation become and what do you remember about it from Calculus I? What if instead $D = 0$?
- (b) By setting $u = \hat{u}U$, $x = \hat{x}X$ and $t = \hat{t}T$, nondimensionalise the model to reduce it to the equation

$$\frac{\partial \hat{u}}{\partial \hat{t}} = \frac{\partial^2 \hat{u}}{\partial \hat{x}^2} + \alpha \hat{u}(1 - \beta \hat{u}), \quad \hat{x} \in [0, 1], \quad (11)$$

stating clearly how the new and old variables are related. (Hint: look at how the domain has changed as well.)

- (c) Without introducing additional parameters (that is, keeping the coefficients of 1 in front of the derivative terms and the same scale for $\hat{x}$), can you find a nondimensionalisation for which we have $\beta = 1$? Can we additionally have $\alpha = 1$? Explain why or why not in each case.

7. Spherical diffusion (Exam section A type)

Consider the diffusion of a bacterial species $c(\mathbf{x}, t)$, $\mathbf{x} \in \mathbb{R}^3$. It satisfies the diffusion equation,

$$\frac{\partial c}{\partial t} = \nabla^2 c, \quad (12)$$

where the diffusion constant has been set to 1. We use a spherical co-ordinate system for which

$$\nabla^2 c = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial c}{\partial r} \right] + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \frac{\partial c}{\partial \theta} \right] + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 c}{\partial \phi^2}. \quad (13)$$

- (a) Write down the diffusion equation for the bacteria, eq. (12), assuming radial symmetry, $c(\mathbf{x}, t) = c(r, t)$.
- (b) Now assume that a spherical region of radius a , centred at $r = 0$, contains a chemical which kills the bacteria in the region. What boundary condition does this imply for this problem?
- (c) Use this boundary condition, as well as $c(r \rightarrow \infty, t) = c_0$, to derive an equilibrium profile of this system.

8. Reflecting and absorbing boundary conditions (Exam section A type)

A bacterial population density c disperses diffusively in a 1D channel between $x = 0$ and $x = L$, with a diffusion constant D (we assume it follows Fick's law). It is advected by the surrounding fluid, moving at a velocity v . Bacteria are reflected at the boundary $x = 0$, while a constant concentration c_0 is maintained at all times at $x = L$.

- (a) Write down the equation for $c(x, t)$.
- (b) Write down the boundary conditions at $x = 0$ and $x = L$ for c .
- (c) Find the equilibrium of this system.

9. Yet more advection–diffusion (Exam section A type)

Consider a one-dimensional concentration of ants, $c(x, t)$, in a channel $x = [0, L]$. Assume that the ant population's bulk motion is governed by the advection–diffusion equation,

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - v \frac{\partial c}{\partial x}. \quad (14)$$

- (a) Derive an expression for the ant flux $F(x, t)$, whose definition comes from comparing to a conservation law in the form

$$\frac{\partial c}{\partial t} = -\frac{\partial F}{\partial x} + f. \quad (15)$$

The flux F basically represents the local change in c accounting for the fact it is diffusing (spreading out) but also moving, both of which cause c to change locally.

- (b) By imposing a radiative boundary condition $F(L, t) = -\gamma$ (concentration points outward) and setting $c(0, t) = 0$, solve for the equilibrium of this system.
- (c) What is the physical interpretation of this distribution in which the diffusion rate is significantly faster than any advective effect, i.e. $vL/D \ll 1$?

10. Bacterial diffusion (Exam section A, August 2017)

Consider a species of bacterium represented by its density (number per unit length) $u(x, t)$ and constrained to a long thin tube (1D domain). The bacteria move subject to diffusion and advection with a constant velocity v . The diffusion follows Fick's law, and the bacteria reproduce at a rate which is proportional to u , with a constant of proportionality λ . The length of the tube is L . At one end (say $x = 0$) there is some chemical which kills the bacteria instantaneously. At the other end of the tube, the bacteria are flowing in with a density gradient μ .

- (a) Explain why the bacteria should be modelled with the equation

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} - v \frac{\partial u}{\partial x} + \lambda u \quad (16)$$

and state the boundary conditions.

- (b) Solve for the equilibrium of the system on the assumption that $\lambda D > v^2$.
- (c) Describe what happens to the solution in the limit $\lambda \rightarrow \infty$ with v and D fixed.

11. Diffusion with a source term

In class, we solved the 1D diffusion equation for a population density $u(x, t)$ on a bounded domain $x \in [0, L]$. Now consider the diffusion equation on the same domain, but with an additional source term which represents exponential growth of the population, equal to βu . The population is constrained to be zero at the boundaries, and the initial condition is a Dirac delta function at $x = L/2$.

- (a) Solve this problem for $u(x, t)$, using the separation of variables method. You should present your final answer as a summation, with all constants reasonably evaluated.

(b) Give a sketch of $u(x, t)$ for a few different t .

12. Two separate Fourier transform practice exercises

(i) Consider the square pulse

$$c(x, 0) = \begin{cases} 0 & x < -a, \\ 1 & -a \leq x \leq a, \\ 0 & x > a. \end{cases} \quad (17)$$

Calculate its Fourier transform.

(ii) Solve the fundamental problem for the 1D advection–diffusion equation using Fick’s law, no source and a velocity v .

13. Fourier transform (Exam section A, 2017)

Consider the following PDE and its fundamental problem:

$$\frac{\partial^2 c_f}{\partial t^2} = D \frac{\partial^2 c_f}{\partial x^2}, \quad c_f(x, 0) = \delta(x), \quad \frac{\partial c_f}{\partial t}(x, 0) = 0, \quad c_f(\pm\infty, t) = 0. \quad (18)$$

Here $\delta(x)$ is the Dirac delta distribution and $D > 0$. We define the Fourier transform as

$$\mathcal{F}[c(x)](k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} c(x) dx, \quad (19)$$

and its inverse as

$$\mathcal{F}^{-1}[c(k)](x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} c(k) dk. \quad (20)$$

(a) Show that eq. (18) implies

$$\mathcal{F}[c_f(x, t)](k) = \frac{1}{\sqrt{2\pi}} \cos(k\sqrt{Dt}). \quad (21)$$

(b) Using a Green’s function, solve the problem

$$\frac{\partial^2 c}{\partial t^2} = D \frac{\partial^2 c}{\partial x^2}, \quad c(x, 0) = g(x), \quad \frac{\partial c}{\partial t}(x, 0) = 0, \quad c(\pm\infty, t) = 0, \quad (22)$$

$$g(x) = \begin{cases} 0 & x < -a, \\ 1 & -a \leq x \leq a, \\ 0 & x > a, \end{cases} \quad (23)$$

and describe briefly the changing shape of the distribution over time. You may use the result

$$\mathcal{F}[\cos(ax)](k) = \sqrt{\frac{\pi}{2}} [\delta(k - a) + \delta(k + a)]. \quad (24)$$

14. Similarity solution (Exam section A, 2018)

Consider the following non-autonomous partial differential equation for a function $c(x, t)$,

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - \frac{1}{t^{1/2}} \frac{\partial c}{\partial x}, \quad (25)$$

where D is a positive constant, $x \in \mathbb{R}$ and $t > 0$.

(a) Assume a similarity solution v which relates to c as

$$c(x, t) = \frac{1}{t^\alpha} v(y), \quad y = x/t^\beta. \quad (26)$$

Show that, for a suitable choice of β , (25) reduces to the following ODE in y ,

$$D \frac{d^2 v}{dy^2} - \frac{dv}{dy} + \alpha v + y\beta \frac{dv}{dy} = 0. \quad (27)$$

(b) Show by integration, that, for a suitable choice of α , this equation can be written as

$$D \frac{dv}{dy} + v \left(\frac{1}{2}y - 1 \right) = C, \quad (28)$$

where C is a constant of integration.

(c) There is a turning point, $\partial c / \partial x = 0$, whose position is $x = 2t^{1/2}$. Assuming α and β take the values found in (a) and (b), show that the solution to eq. (25) takes the form

$$c(x, t) = t^{-1/2} A \exp \left(\frac{x(1 - \frac{1}{4t^{1/2}}x)}{Dt^{1/2}} \right), \quad (29)$$

where A is a constant of integration.